Comparing Risk Models for GPU Kernel Replacement: Balanced Evidence Across Operating Budgets

WANG Zirui
ziruiwang@link.cuhk.edu.hk

ZHANG Juncheng
jczhang25@cse.cuhk.edu.hk

Abstract

GPU kernel replacement is commonly screened by comparing kernel outputs against a numerical tolerance, but local agreement does not ensure task-level agreement. We fit twelve risk models on 6,928 grouped perturbation observations and transfer their predictions to 2,739 real replacements. Scores use shared balanced replacement draws across 21 risk-budget pairs. Mean F1 ranges from 0.4867 for the prevalence baseline to 0.8484 for weighted k-nearest neighbors (KNN); the grouped multilayer perceptron (MLP) and multi-output random forest score 0.8460 and 0.8411. Rankings vary across budgets and metrics, while the leading mean F1 scores are close relative to within-budget repeat variation. The study characterizes screening trade-offs across model families without identifying a universally best model.

1 Introduction

Modern model-serving systems rely on highly optimized GPU kernels. Replacing a reference implementation with a faster kernel can improve throughput, but changes in reduction order, fused arithmetic, or intermediate precision can create floating-point discrepancies. Numerical analysis explains how such errors accumulate through computation graphs [1, 2, 3]. Recent inference studies also report output changes under different execution settings and low-precision arithmetic [4, 5].

Kernel-generation benchmarks make local correctness checks practical by comparing candidate-kernel outputs with a reference under a numerical tolerance [6, 7]. A passing check establishes that the compared tensors are close on the probe inputs. It does not establish that a replacement preserves the model's final class, answer, transcript, or correctness decision. The same numerical discrepancy can have different task-level effects depending on the affected operator and inference context.

We test whether risk models trained on controlled perturbations transfer to real replacements. The first endpoint is an *output change*: the task-level discrete output differs from the reference. The second is a *negative flip*: a request that the reference execution answers correctly becomes incorrect under the replacement. A negative flip is a subset of output changes and also depends on reference correctness. We compare the estimated rates of these outcomes because they capture different consequences of the same numerical discrepancy.

We use the supplied Kernel-Correctness study's 13-category operator-role taxonomy and balanced replacement protocol [8]. Each model maps maximum local discrepancy, task scope, dtype, and operator role to estimated output-change and negative-flip rates. We compare twelve fixed models spanning prevalence, monotone regression, linear and spline models, kernel regression, nearest neighbors, a neural network, single trees, and tree ensembles. Models are frozen before replacement outcomes are scored, and all are evaluated on the same balanced replacement draws. The compar-

CSCI 5150 Course Project.

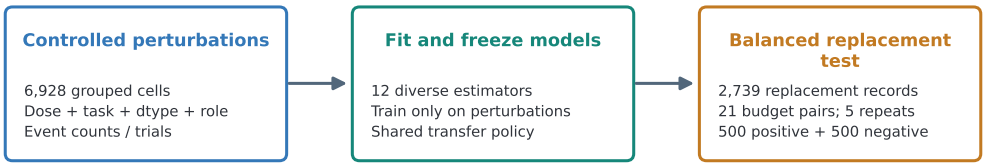


Figure 1: Models are fit on grouped controlled-perturbation observations and applied to real replacements using maximum local discrepancy. Replacement outcomes are withheld from model fitting and selection.

ison asks how model family affects average screening performance, how rankings change across risk-budget pairs, and which models retain recall on replacements with larger net accuracy drops.

2 Related Work

Numerical variation in model inference. Floating-point behavior depends on arithmetic precision, evaluation order, and accumulated rounding error [1]. Accuracy-aware compilation and hardware emulation study how numerical deviations can be bounded or exposed before deployment [2, 3]. FlashAttention-3 uses low precision and asynchronous execution for speed [5], while Yuan et al. [4] show that execution settings can alter language-model outputs. The supplied Kernel-Correctness study connects controlled perturbations to measured replacement outcomes and specifies the role taxonomy and balanced evaluation protocol used here [8].

Kernel generation and local correctness. KernelBench evaluates generated GPU kernels for efficiency and local correctness [6]. TritonBench and MultiKernelBench extend evaluation to Triton operators and additional platforms [7, 9]. We use the local discrepancy measured in this stage as a feature for predicting task-level outcomes after replacement.

Risk estimation. Isotonic regression provides a nonparametric estimate constrained to be monotone in discrepancy [10]. The comparison also includes regression and spline models [11], support-vector regression [12], nearest-neighbor methods [13], multilayer networks [14], and tree ensembles [15, 16, 17, 18]. These methods encode different assumptions about smoothness, locality, partitions, and ensemble construction. Work on preserving sample-level correctness across model updates provides a related correctness endpoint [19]. Here, model parameters remain fixed while the kernel implementation changes.

3 Methodology

Figure 1 summarizes the perturbation-to-replacement protocol. We adopt the source study's operator-role taxonomy and balanced sampling design [8].

3.1 Task and grouped observations

Let $f_{\mathrm{ref}}(x)$ be the final output of a reference model execution on request x, and let $f_{\mathrm{alt}}(x)$ be the output after perturbing or replacing one kernel. Let $C(o, y) \in \{0, 1\}$ indicate correctness against the task target y. We define

$$Z_{\mathrm{change}} = \mathbf{1}\{f_{\mathrm{alt}}(x) \neq f_{\mathrm{ref}}(x)\}, \qquad Z_{\mathrm{flip}} = \mathbf{1}\{C(f_{\mathrm{ref}}(x), y) = 1 \wedge C(f_{\mathrm{alt}}(x), y) = 0\}. \quad (1)$$

The output comparison uses the task's discrete result: top-1 label for classification, selected answer for multiple-choice QA, parsed answer for GSM8K, extracted span for extractive QA, and decoded transcript for ASR. Negative flip uses the reference-correct-to-alternative-wrong transition.

The two endpoints are nested but not interchangeable. An output change can leave task correctness unchanged, whereas a negative flip records a safety-critical change from correct to incorrect. We therefore fit both rates and retain the ordering constraint $p_{\text{flip}} \leq p_{\text{change}}$ after prediction. The joint screening label uses both endpoints, while the net-accuracy-drop bands provide a separate diagnostic for the magnitude of correctness loss. This separation lets the comparison distinguish broad behavioral sensitivity from changes that directly harm task accuracy.

The perturbation corpus is represented as grouped-binomial observations [20]. We define a context by its model, logical kernel, dtype, and benchmark. A cell adds one perturbation setting. For endpoint e, cell i contains $k_{i,e}$ events among n_i requests, with empirical rate $r_{i,e} = k_{i,e}/n_i$. We retain event and trial counts rather than expanding rates into pseudo-independent rows. The training snapshot has 6,928 cells and 1,299 contexts spanning 13 models, seven task scopes, and FP32, FP16, and BF16.

3.2 Inputs and discrepancy mapping

The shared feature set uses four inputs: $\log_{10}(d)$, task scope, dtype, and one of 13 operator-role categories. The role taxonomy separates attention linear, attention arithmetic, attention PV, attention softmax, normalization, activation, convolution, pooling, MLP projection, generic projection, output head, logit scale, and residual scale. Model identity, candidate identity, and benchmark identity remain available for grouping and diagnostics but are not predictor features.

For a perturbation cell, d is the upper endpoint of its applied magnitude range, denoted `range_high`. For a real replacement, we use the maximum absolute difference between matched reference and optimized kernel outputs, denoted `max_abs_diff`. This max-to-max mapping preserves the upper-bound interpretation of the perturbation dose. An exact-zero replacement discrepancy receives predicted risk zero. A nonzero value outside role-specific support is evaluated at the nearest supported boundary, with fallback from task–dtype–role to task scope and then to pooled training support.

3.3 Compared models

We compare twelve models with fixed settings. The prevalence baseline predicts request-weighted training prevalence. Dose-only isotonic regression uses weighted pooled adjacent violators and smoothed rate $(s + 0.5)/(n + 1)$. Ridge regression uses $\alpha = 1$. Softmax spline logistic regression uses a spline basis of log-dose with task, dtype, and role interactions. RBF support-vector regression uses $C = 1$ and $\epsilon = 0.02$. Weighted KNN averages the 25 closest cosine-distance observations, weighted by trial count and inverse distance. The grouped MLP has two hidden layers (64 and 32 units) and minimizes trial-weighted grouped categorical cross-entropy for the nested outcomes. A decision tree is restricted to depth 2. ExtraTrees uses 120 randomized trees, depth 8, and leaf size 6. Histogram gradient boosting uses 180 iterations and at most 15 leaves. CatBoost uses CPU MultiRMSE with 220 iterations, depth 4, learning rate 0.05, and L2 regularization 5.

The multi-output random forest contains 120 trees with maximum depth 8 and minimum leaf size 6. Each split considers every available feature, and the random seed is 20260905. The forest jointly regresses the output-change and negative-flip rates. During fitting, we scale the negative-flip target by two and restore it to its original scale at prediction. A shared post-fit projection enforces $p_{\text{flip}} \leq p_{\text{change}}$.

All learned predictions are clipped to $[0, 1]$ and projected to satisfy $p_{\text{flip}} \leq p_{\text{change}}$. To apply one consistent transfer policy, each context is evaluated on a 61-point geometric dose grid. We replace raw predictions by their nondecreasing upper envelope and use an upper-step lookup for the observed replacement discrepancy. This post-fit policy makes the monotonicity and support behavior explicit for every model.

The model matrix covers four useful inductive regimes. Prevalence and dose-only isotonic regression provide constant and monotone references. Ridge, softmax spline logistic regression, and RBF support-vector regression impose smooth functional structure. Weighted KNN and the grouped MLP represent local interpolation and learned nonlinear interactions. The depth-2 decision tree and

four tree ensembles represent partition-based rules at different ensemble capacities. This grouping is descriptive rather than a second experimental factor: all models use the same features, transfer policy, and evaluation draws, but their hyperparameters are fixed independently.

3.4 Balanced replacement evaluation

The replacement pool contains 2,739 unique records from the current packaged snapshot. We fit each model on all perturbation cells, save predictions from replacement features, and then score against replacement event counts. A risk-budget pair is $(b_{\text{change}}, b_{\text{flip}})$, whose components set the maximum acceptable rates for the two endpoints. A model flags a replacement if either estimated rate exceeds its budget. The joint-positive label uses the same rule with measured rates. We evaluate 21 pairs formed from 0.1%, 0.2%, 0.5%, 1%, 2%, and 5%, retaining pairs where the negative-flip budget is no larger than the output-change budget.

For replacement record i, let $(\hat{p}_{i,\text{change}}, \hat{p}_{i,\text{flip}})$ be the transferred predictions and let $(\tilde{p}_{i,\text{change}}, \tilde{p}_{i,\text{flip}})$ be the measured replacement rates. The model and observed joint labels for budget pair b are

$$\begin{aligned} \hat{y}_i(b) &= \mathbf{1}\{\hat{p}_{i,\text{change}} > b_{\text{change}} \ \vee \ \hat{p}_{i,\text{flip}} > b_{\text{flip}}\}, \\ y_i(b) &= \mathbf{1}\{\tilde{p}_{i,\text{change}} > b_{\text{change}} \ \vee \ \tilde{p}_{i,\text{flip}} > b_{\text{flip}}\}. \end{aligned} \tag{2}$$

The same rule defines the prediction and the label, so changing a budget changes the evaluation target as well as the decision threshold. Predictions remain frozen throughout the sweep; models are never refit for a particular pair.

For every pair, we sample 500 positive and 500 negative records with replacement. Within each class and net-accuracy-drop band, contexts receive equal total weight. The bands are at most 0.5%, above 0.5% through 1%, and above 1%. Net accuracy drop is the correct-to-wrong count minus the wrong-to-correct count. We repeat each pair five times and reuse the draw manifest across models, for 105,000 draw positions. F1 is averaged across pairs and repeats. Its reported standard deviation is the mean of the 21 within-pair sample standard deviations across the five repeats. Overlapping draws describe resampling variability, not independent-record uncertainty. Balanced scores measure screening discrimination and do not estimate natural replacement prevalence. We inherit master seed 9 from the reference protocol, which reports selecting it retrospectively from 128 candidates [8]. Our run does not search over seeds, so these standard deviations omit uncertainty from that earlier seed selection.

For a sampled record, $y = 1$ denotes a joint-positive replacement and $\hat{y} = 1$ denotes a model flag. Precision is $TP/(TP + FP)$, joint-positive recall is $TP/(TP + FN)$, and F_1 is their harmonic mean [21]. The accepted set consists of records with $\hat{y} = 0$. Its accepted violation fraction is $FN/(FN + TN)$, the share of accepted records that are positive. We also report recall among replacements whose net accuracy drop exceeds 0.5% and 1%. All measures are computed per draw and summarized over repeats and risk-budget pairs. Each draw has equal positive and negative counts, so precision is specific to this balanced design.

The metrics use distinct denominators because screening asks two different questions: how many risky replacements are caught, and how safe the accepted set is. Table 1 summarizes these denominators. Reporting the accepted fraction with its violation fraction is particularly important: a system that flags nearly everything can leave a small accepted set with few violations, but it may provide little useful screening capacity. The severity recalls use only the corresponding high-drop records in each draw, so they measure targeted coverage rather than overall classification quality.

4 Experiments

4.1 Experimental setup

The run uses the current normalized project snapshot and no new GPU execution. The perturbation input contains 6,928 cells from 1,299 contexts; the replacement feature table contains 2,739 records across the same context identities. Thus, the experiment tests transfer from controlled perturbations to real replacements within represented contexts, not generalization to unseen contexts. Replacement outcomes remain withheld from fitting and model selection. The implementation uses NumPy 2.1.3, pandas 2.2.3, and scikit-learn 1.6.1. Tree and CatBoost models run on CPU. Every model uses the same operator-role mapping, dose mapping, support policy, and balanced replacement draws.

Table 1: Interpretation of the screening metrics. All entries are computed on the balanced replacement draws.

Metric	Denominator	Question answered
Precision	Flagged replacements	How many flags are joint-positive?
Joint-positive recall	Joint-positive replacements	How many risky replacements are flagged?
Accepted violation fraction	Accepted replacements	How many accepted replacements are positive?
Accepted fraction	All sampled replacements	How much of the pool passes screening?
Severity recall	Replacements above a drop threshold	How many larger-drop cases are flagged?

Table 2: Balanced replacement screening across twelve risk models. F1 is the mean across 21 risk-budget pairs and five repeats. The reported SD is the mean within-pair sample SD across repeats. All models use the same balanced replacement draws.

Model	F1 ↑ (mean ± SD)	Precision ↑	Joint-positive recall ↑	$> 0.5\%$ recall ↑	$> 1\%$ recall ↑	Accepted violation fraction ↓
Prevalence baseline	0.4867 ± 0.0066	0.4097	0.6000	0.6388	0.6497	0.2879
Dose-only isotonic regression	0.7468 ± 0.0104	0.6712	0.8473	0.8521	0.8480	0.2118
Ridge regression	0.8075 ± 0.0113	0.7490	0.8801	0.8154	0.7105	0.1445
Softmax spline logistic regression	0.8338 ± 0.0116	0.7954	0.8784	0.8239	0.7267	0.1353
RBF support-vector regression	0.8020 ± 0.0103	0.7386	0.8827	0.8411	0.7624	0.1444
Weighted KNN	0.8484 ± 0.0112	0.8311	0.8678	0.8042	0.6951	0.1373
Grouped MLP	0.8460 ± 0.0108	0.8261	0.8686	0.7977	0.6770	0.1381
Decision tree (depth 2)	0.7897 ± 0.0119	0.7860	0.8168	0.8506	0.7905	0.1886
Multi-output random forest	0.8411 ± 0.0115	0.8220	0.8634	0.8375	0.7638	0.1419
ExtraTrees	0.8384 ± 0.0113	0.8164	0.8638	0.8274	0.7453	0.1429
Histogram gradient boosting	0.8384 ± 0.0117	0.8098	0.8710	0.8302	0.7433	0.1385
CatBoost	0.8346 ± 0.0119	0.8017	0.8727	0.8409	0.7626	0.1392

The replacement pool contains 121 records with net accuracy drop above 0.5% and 64 above 1%. These cases are uncommon in the full pool, so the balanced protocol preserves them as explicit severity strata instead of allowing the largest class to determine recall. Within each class and severity band, context identifiers receive equal total weight during resampling. The resulting score answers how a model behaves when each selected risk group is represented in the evaluation, rather than how often that group would appear under the natural replacement distribution.

4.2 Main comparison

Table 2 reports mean performance across 21 risk-budget pairs and five fixed repetitions. A replacement is joint-positive if either measured endpoint rate exceeds its corresponding budget. The two severity columns report recall among replacements whose net accuracy drop exceeds 0.5% and 1%. Accepted violation fraction is the share of accepted replacements that are joint-positive.

Weighted KNN has the highest mean F1 (0.8484), followed by the grouped MLP (0.8460) and multi-output random forest (0.8411). Their mean within-pair repeat standard deviations are 0.0112, 0.0108, and 0.0115. The gaps between their mean F1 values are smaller than these repeat-level standard deviations, so the average ordering is descriptive. Other criteria favor different models: CatBoost has the highest joint-positive recall (0.8727), softmax spline logistic regression the lowest accepted violation fraction (0.1353), and the depth-2 decision tree the highest recall in both net-accuracy-drop groups (0.8506 and 0.7905).

Mean F1 rises from 0.4867 for the prevalence baseline to 0.7468 for dose-only isotonic regression and ranges from 0.7897 to 0.8484 for models that also use context features. This pattern is consistent with context features helping to distinguish replacements, but it does not isolate their effect: the models also differ in functional form, regularization, and optimization.

Accepted-set size gives the denominator needed to interpret the violation metric. The prevalence baseline accepts 0.5118 of sampled records, while dose-only isotonic regression accepts 0.3631. The leading F1 models accept similar shares: 0.4771 for weighted KNN, 0.4730 for the grouped MLP, and 0.4738 for the multi-output random forest. Softmax spline logistic regression has the lowest accepted violation fraction, 0.1353, while accepting 0.4460 of records. The shallow tree accepts 0.4663 but leaves 0.1886 of its accepted records positive. Thus, a lower violation fraction

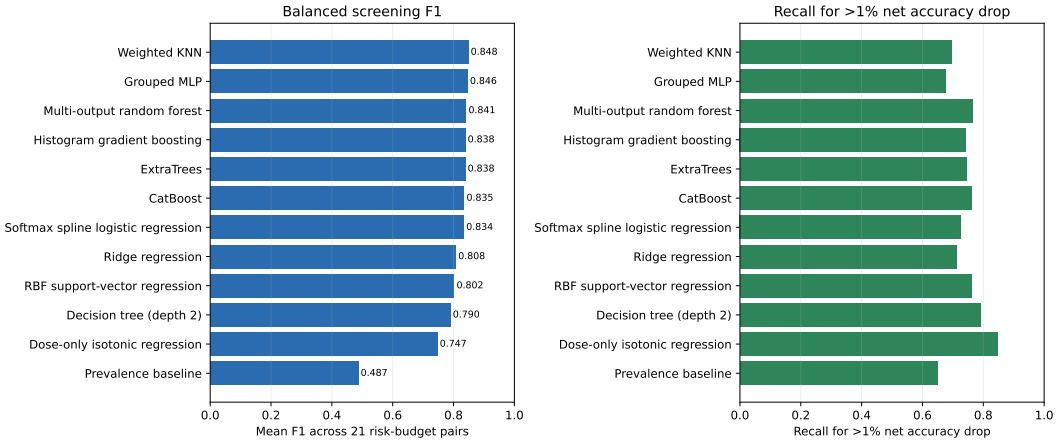


Figure 2: Mean F1 and recall among replacements with net accuracy drop above 1% across the twelve risk models. Both panels use the same balanced replacement draws.

can reflect a safer accepted set, a smaller accepted set, or both; the accepted fraction should be reported with it.

4.3 Interpreting balanced results

Each risk-budget pair is evaluated on draws with equal positive and negative counts. F1 and precision therefore describe balanced screening rather than performance at natural replacement prevalence. They do not estimate production precision, workload, or calibration. The shared draw manifest pairs model comparisons at the sampled-record level, but the summary reports no paired confidence intervals. Accepted violation fraction is conditioned on records the model accepts, whereas joint-positive recall is conditioned on all positive records. These metrics answer different questions and are not complements.

4.4 Sensitivity to risk-budget pairs

The mean over risk-budget pairs hides variation in the evaluation target. Table 3 summarizes each model's F1 range, standard deviation across pair-level means, and number of pairs ranked first. These quantities measure sensitivity to budget choice, whereas the five-repeat standard deviation in Table 2 measures draw-to-draw variation for a fixed pair.

Weighted KNN has the smallest across-budget standard deviation (0.0174), followed by ExtraTrees (0.0200) and the multi-output random forest (0.0229). KNN ranks first in 12 pairs and the grouped MLP in six; the grouped MLP also has the highest pair-level mean F1 (0.8957). The prevalence baseline varies most because its constant risk estimate sometimes falls below both budgets, leaving it with no positive predictions and zero F1.

The family-level pattern is consistent with different ways of using context. KNN is both accurate on average and stable across budget pairs, which indicates that nearby feature configurations preserve useful local structure under this transfer mapping. The grouped MLP reaches the highest pair-level mean but varies more across budgets, so its nonlinear interactions are sensitive to the chosen operating point. The multi-output random forest, ExtraTrees, and histogram gradient boosting cluster near 0.838–0.841 mean F1 while retaining joint-positive recall between 0.8634 and 0.8710. These observations describe the tested implementations; they do not establish that locality, neural capacity, or tree ensembles is the causal source of a model's advantage.

The severity columns reorder the models because a high overall score does not guarantee coverage of the largest regressions. The depth-2 tree recalls 0.7905 of replacements with net accuracy drop above 1%, compared with 0.7638 for the multi-output random forest, 0.7626 for CatBoost, and 0.6951 for KNN. Its joint-positive recall is lower at 0.8168, and its accepted violation fraction is higher at 0.1886. In this comparison, the tree captures more of the severe tail while leaving a less

Table 3: F1 sensitivity across the 21 risk-budget pairs. Range endpoints and standard deviation are computed over pair-level means. First-place counts use the highest pair mean, with no ties in this run.

Model	F1 range	SD across budgets	First-place pairs
Prevalence baseline	0.000–0.765	0.3530	0
Dose-only isotonic regression	0.703–0.800	0.0293	0
Ridge regression	0.765–0.864	0.0261	0
Softmax spline logistic regression	0.790–0.874	0.0271	1
RBF support-vector regression	0.770–0.870	0.0237	1
Weighted KNN	0.817–0.871	0.0174	12
Grouped MLP	0.800–0.896	0.0263	6
Decision tree (depth 2)	0.711–0.844	0.0439	0
Multi-output random forest	0.808–0.883	0.0229	1
ExtraTrees	0.802–0.874	0.0200	0
Histogram gradient boosting	0.786–0.873	0.0237	0
CatBoost	0.776–0.876	0.0305	0

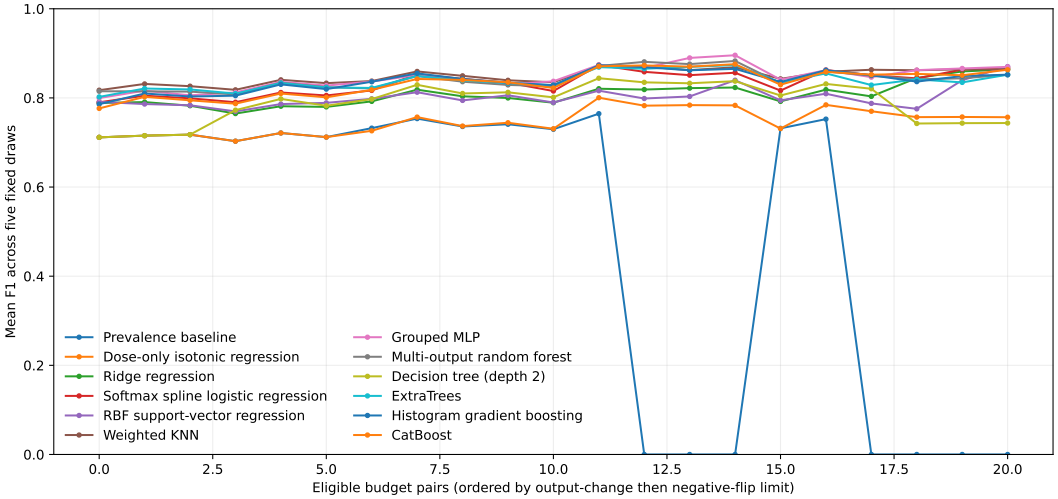


Figure 3: Mean balanced F1 across the 21 output-change/negative-flip risk-budget pairs. Predictions remain frozen while the joint-positive label changes with the budgets.

reliable accepted set. The $> 1\%$ group contains only 64 unique replacements, so the result identifies a useful difference in this snapshot rather than a stable population ranking.

4.5 Budget sensitivity and transfer support

Figure 3 shows F1 across the 21 risk-budget pairs. Model rankings change with the acceptable endpoint rates. Predictions remain frozen, so the curves reflect changes in budget-defined labels rather than refitting.

Changing the output-change budget can alter the positive labels even when the negative-flip budget stays fixed. Because each pair receives equal weight in the average, the aggregate scores may not reflect the risk tolerance of a particular deployment. A deployment decision should therefore use the relevant budget-specific confusion matrix and review cost.

The transfer policy assigns 758 records to the exact-zero state, 1,308 within role-specific dose support, 110 below support, and 563 above support. The 673 below- or above-support records receive predictions through boundary clipping or fallback and therefore have weaker dose support than in-range records.

4.6 Using the comparison to choose a screening policy

The results support a policy-first selection procedure. A team that values balanced discrimination and stable behavior across budgets can start with weighted KNN, which has the highest mean F1 and the smallest across-budget standard deviation. A team that prioritizes catching any joint-positive replacement should inspect CatBoost or histogram gradient boosting because they achieve the highest joint-positive recall among the compared models. A team that treats an unsafe accepted replacement as the main cost should examine the softmax spline logistic model together with the accepted fraction, rather than selecting by F1 alone. If the main concern is a large task-level accuracy loss, the depth-2 tree provides the strongest tail recall in this snapshot, but its lower joint-positive recall and higher accepted violation fraction make the trade-off explicit.

The transfer support status should be part of the same decision. Of the 2,739 replacement records, 758 have zero discrepancy, 1,308 fall within role-specific dose support, and 673 lie below or above that support. Predictions for the last group are produced by boundary clipping or hierarchical fallback. They are useful for ranking candidates under the fixed protocol, but they should not be read as calibrated probabilities without additional data. A practical deployment report should therefore include the selected budget pair, the confusion matrix, accepted fraction, severity recall, and the fraction of records evaluated outside direct support. This reporting format preserves the comparison's empirical scope while making the operational objective explicit.

4.7 Limitations

The results cover one packaged snapshot with 13 models, seven task scopes, 1,299 contexts, and 2,739 replacements. Perturbation and replacement records share context identities, so this evaluation tests transfer between discrepancy-generating mechanisms within represented contexts, not generalization to unseen contexts. Maximum absolute discrepancy omits error sign, distribution, correlation, and accumulation across invocations. Upper-tail labels are sparse, sampling with replacement can repeat high-risk records, and boundary clipping does not guarantee calibrated estimates outside training support. The replacement labels also differ from some historical validation snapshots in the source study.

Model settings are fixed rather than selected through nested context-grouped validation, and compute budgets differ across model families. The comparison therefore measures these implementations, not the best tuned version of each family. The shared operator-role taxonomy is not evaluated through a within-model feature ablation. Balanced draws also do not estimate natural-prevalence precision, calibration, or deployment cost. The 105,000 draw positions overlap, and reported standard deviations omit uncertainty from seed selection, the replacement snapshot, and retraining. A stronger evaluation would use an independently collected replacement pool, paired context-level intervals, and calibration measurements at the natural event rate.

5 Conclusion

We compared twelve risk models under a shared perturbation-to-replacement protocol and balanced replacement evaluation. Weighted KNN and the grouped MLP have the highest mean F1, but rankings shift across risk-budget pairs and metrics. CatBoost has the highest joint-positive recall, softmax spline logistic regression the lowest accepted violation fraction, and the shallow tree the highest recall among larger accuracy-drop cases. The leading mean F1 scores are close relative to repeat-level variation, so the evidence supports a metric- and budget-specific comparison rather than a universal ranking. These conclusions apply to contexts and discrepancy support represented in the current snapshot.

Acknowledgments and AI-use disclosure

OpenAI Codex assisted with analysis-code development and manuscript editing [22]. The authors are responsible for the submitted report.

References

[1] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, second edition, 2002. ISBN 978-0-89871-521-7. doi: 10.1137/1.9780898718027. URL https://doi.org/10.1137/1.9780898718027.

[2] Hashim Sharif, Prakalp Srivastava, Muhammad Huzaifa, Maria Kotsifakou, Keyur Joshi, Yasmin Sarita, Nathan Zhao, Vikram S. Adve, Sasa Misailovic, and Sarita Adve. ApproxHPVM: A portable compiler IR for accuracy-aware optimizations. *Proceedings of the ACM on Programming Languages*, 3(OOPSLA), 2019. doi: 10.1145/3360612. URL https://doi.org/10.1145/3360612.

[3] Chenhao Jiang, Anand Jayarajan, Hao Lu, and Gennady Pekhimenko. Arbitor: A numerically accurate hardware emulation tool for DNN accelerators. In *2023 USENIX Annual Technical Conference (USENIX ATC 23)*, pages 519–536, Boston, MA, 2023. USENIX Association. URL https://www.usenix.org/conference/atc23/presentation/jiang-chenhao.

[4] Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. Understanding and mitigating numerical sources of nondeterminism in LLM inference, 2025. URL https://arxiv.org/abs/2506.09501.

[5] Jay Shah, Ganesh Bikshandi, Ying Zhang, Vijay Thakkar, Pradeep Ramani, and Tri Dao. FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://arxiv.org/abs/2407.08608.

[6] Anne Ouyang, Simon Guo, Simran Arora, Alex L. Zhang, William Hu, Christopher Ré, and Azalia Mirhoseini. KernelBench: Can LLMs write efficient GPU kernels?, 2025. URL https://arxiv.org/abs/2502.10517.

[7] Jianling Li, Shangzhan Li, Zhenye Gao, Qi Shi, Yuxuan Li, Zefan Wang, Jiacheng Huang, Haojie Wang, Jianrong Wang, Xu Han, Zhiyuan Liu, and Maosong Sun. TritonBench: Benchmarking large language model capabilities for generating Triton operators, 2025. URL https://arxiv.org/abs/2502.14752.

[8] Anonymous Author(s). Understanding Tolerance Threshold Setup in Automated GPU Kernel Optimization. Unpublished manuscript used as the reference protocol for this course project, 2026.

[9] Zhongzhen Wen, Yinghui Zhang, Zhong Li, Zhongxin Liu, Linna Xie, and Tian Zhang. MultiKernelBench: A multi-platform benchmark for kernel generation, 2025. URL https://arxiv.org/abs/2507.17773.

[10] Alexander Henzi, Johanna F. Ziegel, and Tilmann Gneiting. Isotonic distributional regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(5):963–993, 2021. doi: 10.1111/rssb.12450. URL https://doi.org/10.1111/rssb.12450.

[11] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics. Springer New York, second edition, 2009. doi: 10.1007/978-0-387-84858-7. URL https://doi.org/10.1007/978-0-387-84858-7.

[12] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995. doi: 10.1007/BF00994018. URL https://doi.org/10.1007/BF00994018.

[13] Thomas M. Cover and Peter E. Hart. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967. doi: 10.1109/TIT.1967.1053964. URL https://doi.org/10.1109/TIT.1967.1053964.

[14] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323 (6088):533–536, 1986. doi: 10.1038/323533a0. URL https://doi.org/10.1038/323533a0.

[15] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324. URL https://doi.org/10.1023/A:1010933404324.

[16] Pierre Geurts, Damien Ernst, and Louis Wehenkel. Extremely randomized trees. *Machine Learning*, 63(1):3–42, 2006. doi: 10.1007/s10994-006-6226-1. URL https://doi.org/10.1007/s10994-006-6226-1.

[17] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451. URL https://doi.org/10.1214/aos/1013203451.

[18] Anna Veronika Dorogush, Vasily Ershov, and Andrey Gulin. CatBoost: Gradient boosting with categorical features support, 2018. URL https://arxiv.org/abs/1810.11363.

[19] Sijie Yan, Yuanjun Xiong, Kaustav Kundu, Shuo Yang, Siqi Deng, Meng Wang, Wei Xia, and Stefano Soatto. Positive-congruent training: Towards regression-free model updates. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021. URL https://arxiv.org/abs/2011.09161.

[20] Peter McCullagh and John A. Nelder. *Generalized Linear Models*. Chapman and Hall, second edition, 1989. ISBN 978-0-412-31760-6.

[21] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009. doi: 10.1016/j.ipm.2009.03.002. URL https://doi.org/10.1016/j.ipm.2009.03.002.

[22] OpenAI. Codex. Software tool, 2025. URL https://github.com/openai/codex.